

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	T.S.P Madhuri
Student Name	V.PRANAV
Roll Number	2520030207
Branch	CSE
Course Outcome	CO-6

Case Study Topic: Coin Change DP — Minimum Number of Coins to Make Change

Work Done Summary: In this case study, we implemented a Dynamic Programming solution to determine the minimum number of coins required to make a target amount. The program stores intermediate results in a DP array and computes the optimal solution efficiently. We also displayed the minimum number of coins required for the given amount.

Implementation Details:

1. Defined the available coin denominations.
2. Created a DP array to store minimum coins required.
3. Initialized DP array with large values.
4. Set $dp[0] = 0$ as the base case.
5. Iterated through all coin denominations.
6. Updated DP values using recurrence relation.
7. Displayed the minimum number of coins required.

Result: Screenshots / Output

```
Coin Change DP
Amount = 43
Minimum Coins Required = 4

Coin Denominations:
1 2 5 10 20
```

The Dynamic Programming approach successfully computed the minimum number of coins required.

Coin Denominations:

1, 2, 5, 10, 20

Target Amount:43

Minimum Coins Required: 4

One Possible Combination:

$20 + 20 + 2 + 1$

Time Complexity:

$O(\text{amount} \times \text{number of coins})$

The DP approach guarantees the optimal solution for any coin set.

GitHub Link: <https://github.com/pranavvv-18/DSA-2-CASE-STUDY>

Student Signature	Faculty Signature